Omkar Pote

omkarpote.com | omkrpt@gmail.com | github.com/omkrpt | linkedin.com/in/omkrpt

Education

Purdue University **Aug 2024 – Dec 2026 (expected)**

M.S. in Computer and Information Technology

Thesis successfully defended (Advisor: Dr. Romila Pradhan)
Historical Segmentation for Budgeted Preprocessing Pipeline Selection under Distribution Shift (link)

Indian Institute of Technology, Madras **Jul 2016 – May 2020**

Bachelor of Technology

Research Experience

Purdue University **Aug 2024 – Dec 2026**

M.S. Thesis Research *Advisor: Dr. Romila Pradhan*

- Developed a method for selecting preprocessing pipelines under distribution shift by reusing pipeline evaluations from historical data regions.
- Used a shared representation of feature distributions, missingness, outliers, and category rarity for historical segmentation and retrieval; fitted regional pipelines with a global fallback.
- Evaluated the approach on Bank Marketing and Diabetes Readmission. For the 96-candidate Diabetes search space, reduced incoming candidate evaluations to three (96.9% fewer), excluding historical construction and final refits.

Purdue University **Aug 2024 – May 2025**

Graduate Research Assistant, Responsible Data Science Lab *Part-Time*

- Worked under Dr. Romila Pradhan on task vector arithmetic for reducing bias in machine learning models; presented the work at CERIAS and Purdue AARC Applied AI Day.
- Conducted exploratory study of large model management and model lakes, focusing on metadata quality, model provenance, versioning, and model discovery.

IIT Madras **Oct 2023 – Jun 2024**

Project Scientist, DART / iACE Labs

- Worked in the American Express Lab for Data Analytics, Risk & Technology (DART) / Intelligent Applications in Chemical Engineering (iACE) Lab under Dr. Rajagopalan Srinivasan.
- Developed model-agnostic counterfactual explanations for multivariate time-series classification, focusing on fault diagnosis and recovery; contributed to the ESCAPE 2025 short paper listed below.

Publications & Presentations

Peer-Reviewed Publications

- **Pote, O.**, Majji, D., Bhakte, A., Srinivasan, B., and Srinivasan, R., *"Timeseries-based Counterfactual Explanations for Process Fault Diagnosis and Recovery."* 35th European Symposium on Computer Aided Process Engineering (ESCAPE), 2025. *Book of Short Papers*, pp. 235–236. Peer-reviewed short paper. (link)

Poster Presentations

- **Pote, O.** and Pradhan, R., *"Unlearning Machine Learning Bias using Task Vector Arithmetic."* (link) 26th CERIAS Annual Cybersecurity Symposium, Purdue University, 2025 – ***Second Place, Poster Competition***
- **Pote, O.** (Advisor: Dr. Romila Pradhan), *"Resolving Machine Learning Bias using Task Vector Arithmetic."* Purdue AARC Applied AI Day, Purdue University, 2025.
- **Pote, O.**, Iturburu, L., Dyke, S., and Pradhan, R., *"Explaining AI-based System Decisions in Structural Engineering."* NSF IUCRC VISER Center Planning Meeting, University of Houston, 2024.

Teaching Experience

Purdue University **Aug 2025 – Dec 2025**

Graduate Teaching Assistant, CNIT 18000: Introduction to Systems Development *Part-Time*

- Independently ran two laboratory sections of CNIT 18000: Introduction to Systems Development, with sole responsibility for lab instruction, grading, and student support.
- Taught systems-development and database concepts, guided students through laboratory assignments, and provided feedback on coursework.

Academic Service

- External Reviewer, International Conference on Extending Database Technology (EDBT), 2026.
- External Reviewer, IEEE International Conference on Data Engineering (ICDE), 2025.

Professional Experience

IIT Madras **Oct 2023 – Jun 2024**

Software Quality Mentor, HILCPS Lab

- Worked with the Human-in-the-Loop Cyber-Physical Systems Lab, led by Dr. Babji Srinivasan. Conducted code reviews and fortnightly mentoring sessions on version control, testing, design patterns, and software development practices.
- Led architecture, planning, and cloud migration for an IoT project using Flutter, Django, and Embedded C.

Capillary Technologies **May 2020 – Sep 2023**

Software Development Engineer, Payments *Dec 2021 – Sep 2023*

- Designed and implemented microservices for cashless fuel payments, integrating payment providers, fuel stations, and third-party services.
- Built services using Spring Boot, MySQL, MongoDB, and RabbitMQ, supporting 10,000+ transactions per minute; established CI/CD pipelines and monitoring.

Big Data Engineer, Extract-Transform-Load (ETL) *Oct 2020 – Nov 2021*

- Managed data warehousing and ETL workflows processing 6+ billion records daily; developed REST APIs and optimized Spark queries on AWS EMR.

Intern *May 2020 – Sep 2020*

- Built tools for controlling and monitoring ETL workflows represented as directed acyclic graphs in Neo4j.

Tesco **May 2019 – Jul 2019**

Software Engineering Intern

- Automated food-donation workflows using a product-classification API and an Apache NiFi / Flask pipeline, supporting processes serving 2,500+ stores and 6,000+ charities.

Crest IT Consulting **Jun 2018 – Jul 2018**

Web Development Intern

- Developed the company website and visual identity as part of a rebranding initiative.

Technical Skills

Programming: Python, SQL, Java, Scala, C/C++, Bash, MATLAB

ML & Data Tools: NumPy, pandas, scikit-learn, PyTorch, Hugging Face Hub API, Sentence Transformers

Data Engineering & Databases: ETL pipelines, Apache Spark, MySQL, MongoDB, Neo4j

Research Methods: Data preprocessing and pipeline selection, metafeature engineering, clustering, cross-validation, evaluation under distribution shift, counterfactual explanations, fairness evaluation, task-vector arithmetic and model-weight editing

Development & Infrastructure: Git, Docker, Kubernetes, AWS (including EMR), CI/CD